



ARTIFICIAL INTELLIGENCE AND REGULATORY SANDBOX (SINGAPORE)

SUBMIT BY : PARISTA DESAR

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INTRODUCTION TO ARTIFICIAL INTELLIGENCE

- - AI refers to systems that perform tasks requiring human intelligence.
- - Includes machine learning, natural language processing, and automation.
- - Impact sectors: finance, healthcare, logistics, governance.

AI IN SINGAPORE

- - Part of Singapore's Smart Nation initiative.
- - Focus on responsible innovation and governance.
- - Government investment in AI R&D and workforce training.
- - National AI Strategy launched to integrate AI into key sectors.

REGULATORY SANDBOX CONCEPT

- - A framework allowing controlled experimentation of innovative tech.
- - Businesses test solutions with relaxed regulations.
- - Authorities observe risks and benefits before full-scale deployment.

SINGAPORE'S AI REGULATORY SANDBOX

- - Managed primarily by the Monetary Authority of Singapore (MAS).
- - Enables fintech firms to test AI-driven tools safely.
- - Supports AI in fraud detection, risk scoring, customer service systems.
- - Collaboration between regulators, AI developers, and industry stakeholders.

BENEFITS OF AI REGULATORY SANDBOXES

- - Promotes innovation without compromising consumer safety.
- - Reduces regulatory barriers for startups.
- - Helps regulators refine AI governance frameworks.
- - Encourages ethical and transparent AI development.

AI GOVERNANCE IN SINGAPORE

- - Model AI Governance Framework provides guidelines for businesses.
- - Focus on fairness, transparency, accountability, and human oversight.
- - Sandbox supports practical implementation of these principles.

CASE EXAMPLES

- - AI-driven credit assessment tools tested within MAS Sandbox.
- - Chatbot-based financial advisory systems.
- - Machine learning models for fraud analytics.

CHALLENGES

- - Balancing innovation and regulation.
- - Ensuring consumer privacy and data protection.
- - Avoiding algorithmic bias.
- - Need for continuous regulatory updates.

FUTURE OF AI AND SANDBOX IN SINGAPORE

- - Expansion of sandbox frameworks to other sectors.
- - Stronger cross-border regulatory collaboration.
- - Increased adoption of explainable and trustworthy AI.
- - Continued evolution of the National AI Strategy.

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